

## 1 First equation

Some introductory text...

**Example 1.1**

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$$f(x) = \frac{x}{1+x^2} \tag{1.1}$$

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### 1.1 More detail

**Example 1.2**

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Here we discuss

$$f(x) = \frac{x+1}{x-1} \tag{1.2}$$

... but don't say anything

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#### 1.1.1 Even more detail

**Example 1.3**

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$$f(x) = \frac{x^2}{1+x^3} \tag{1.3}$$

$$f(x+\delta x) = \frac{(x+\delta x)^2}{1+(x+\delta x)^3} \tag{1.4}$$

$$f(x+\delta x) = \frac{(x+\delta x)^0}{1+(x+\delta x)^3} \tag{1.5}$$

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## 2 Third equation

**Example 2.1**

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The following function...

$$f_1(x) = \frac{x+1}{x-1} \tag{2.1}$$

..is a function

$$f_2(x) = \frac{x^2}{1+x^3} \tag{2.2}$$

$$f_3(x) = \frac{3+x^2}{1-x^3} \tag{2.3}$$

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